

Mobile Testing with Appium + Playwright TypeScript



Automated Testing



<https://appium.io/>

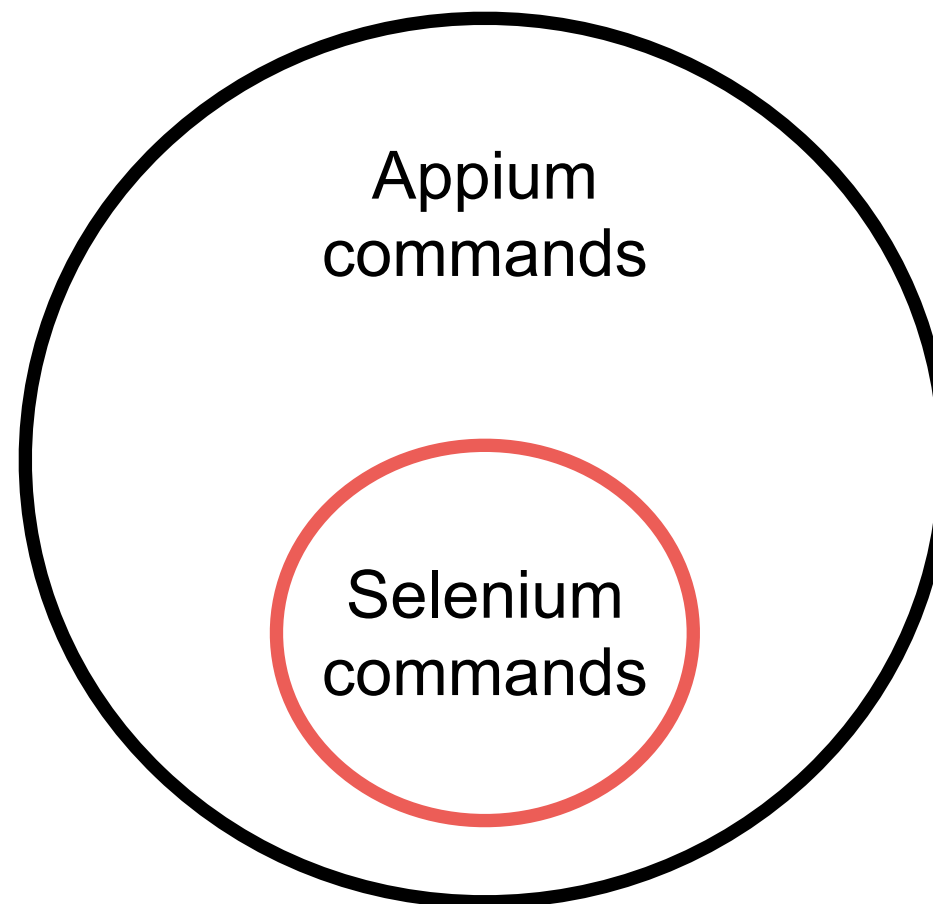


Automated Testing

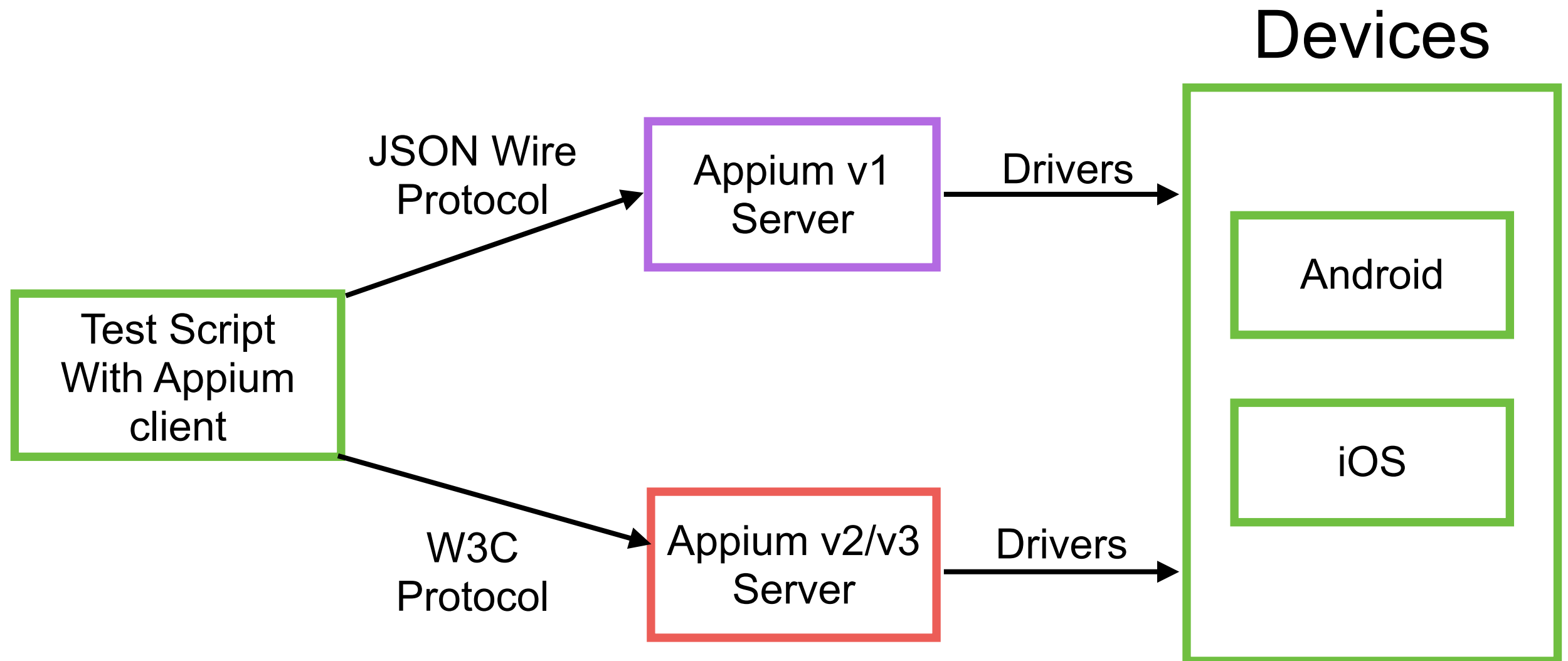


Appium

Automation tool for mobile app
Selenium for mobile
High compatibility with selenium



Architecture



Appium Tools

Appium Server via npm
Appium Inspector

Android

iOS

Web

Desktop

<https://appium.io/docs/en/latest/quickstart/>



Appium Drivers

Platform	Driver	Platform Versions	Appium Version
iOS	XCUITest	9.3+	1.6.0+
	UIAutomation	8.0 to 9.3	All
Android	Espresso	?+	1.9.0+
	UiAutomator2	?+	1.6.0+
	UiAutomator	4.3+	All
Mac	Mac	?+	1.6.4+
Windows	Windows	10+	1.6.0+

<https://appium.io/docs/en/latest/ecosystem/drivers/>



Appium Clients

JavaScript/TypeScript

Python

Java

C#

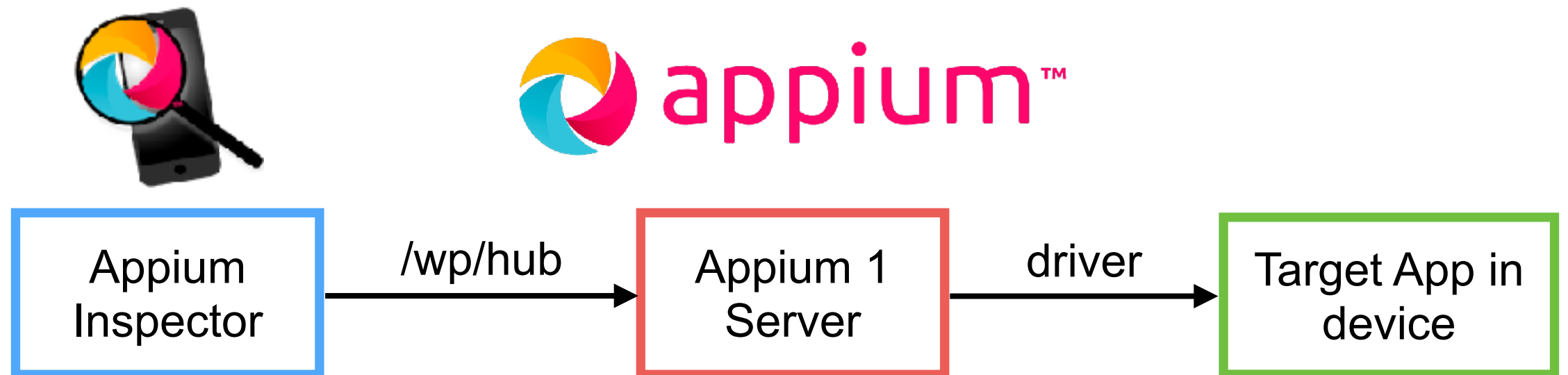
Robotframework

<https://appium.io/docs/en/latest/ecosystem/clients/>



Let's start with Appium

Appium doctor (android, iOS)



Steps to run

Appium doctor and drivers

Start server

Provide target apps (api, ipa/app)

Appium Inspector



Check environments

Android	iOS
APPIUM_HOME	APPIUM_HOME
ANDROID_HOME	Xcode
JAVA_HOME	Xcode command line tools
adb, emulator, apkanalyzer	ios-deploy, applesimutils
NodeJS	NodeJS



List of android's devices

\$adb devices



List of iOS's devices

\$xcrun xctrace list devices

```
== Simulators ==  
Apple TV Simulator (15.2) (9666637B-AD71-47CD-932D-DE7BA9096F46)  
Apple TV 4K (2nd generation) Simulator (15.2) (2B6A15B3-4A03-41B5-B6C6-D7727DCA94D8)  
Apple TV 4K (at 1080p) (2nd generation) Simulator (15.2) (EC5B5310-9DE7-4CB2-8AE2-172FC885DB8C)  
iPad (9th generation) Simulator (15.2) (03B84395-1A2B-4759-B01A-171CD4F8F796)  
iPad Air (4th generation) Simulator (15.2) (6389A3CA-7805-452D-9110-5B3796A9D9BB)  
iPad Pro (11-inch) (3rd generation) Simulator (15.2) (40E522A5-14CE-453C-A54C-29D096F19236)  
iPad Pro (12.9-inch) (5th generation) Simulator (15.2) (1E6E63EF-FA56-4E31-8082-13B0C664AD33)  
iPad Pro (9.7-inch) Simulator (15.2) (9C7181D9-73D6-48BF-8972-80A73ADB2EA9)  
iPad mini (6th generation) Simulator (15.2) (01F8F0AB-1AF5-4FBE-9587-B7713F7BBC6F)  
iPhone 11 Simulator (15.2) (1E75E325-57EC-4D6D-85BF-9958B0A9B158)  
iPhone 11 Pro Simulator (15.2) (7D50F81C-3521-447F-A79C-8613C6C58FEF)  
iPhone 11 Pro Max Simulator (15.2) (FB336CEE-A241-43D1-8704-B0632EA9FD28)  
iPhone 12 Simulator (15.2) (5325CF3D-9576-462A-A110-49A5D76665FE)
```



List of Appium driver

\$appium driver list

```
✓ Listing available drivers (rerun with --verbose for more info)
- uiautomator2@7.0.0 [installed (npm)]
- xcuitest [not installed]
- espresso [not installed]
- mac2 [not installed]
- windows [not installed]
- safari [not installed]
- gecko [not installed]
- chromium [not installed]
```

<https://appium.io/docs/en/latest/ecosystem/drivers/>



Appium Server



Appium Server

Install via npm (required Node.js)

```
$npm install -g appium  
$appium
```

<https://github.com/appium/appium>

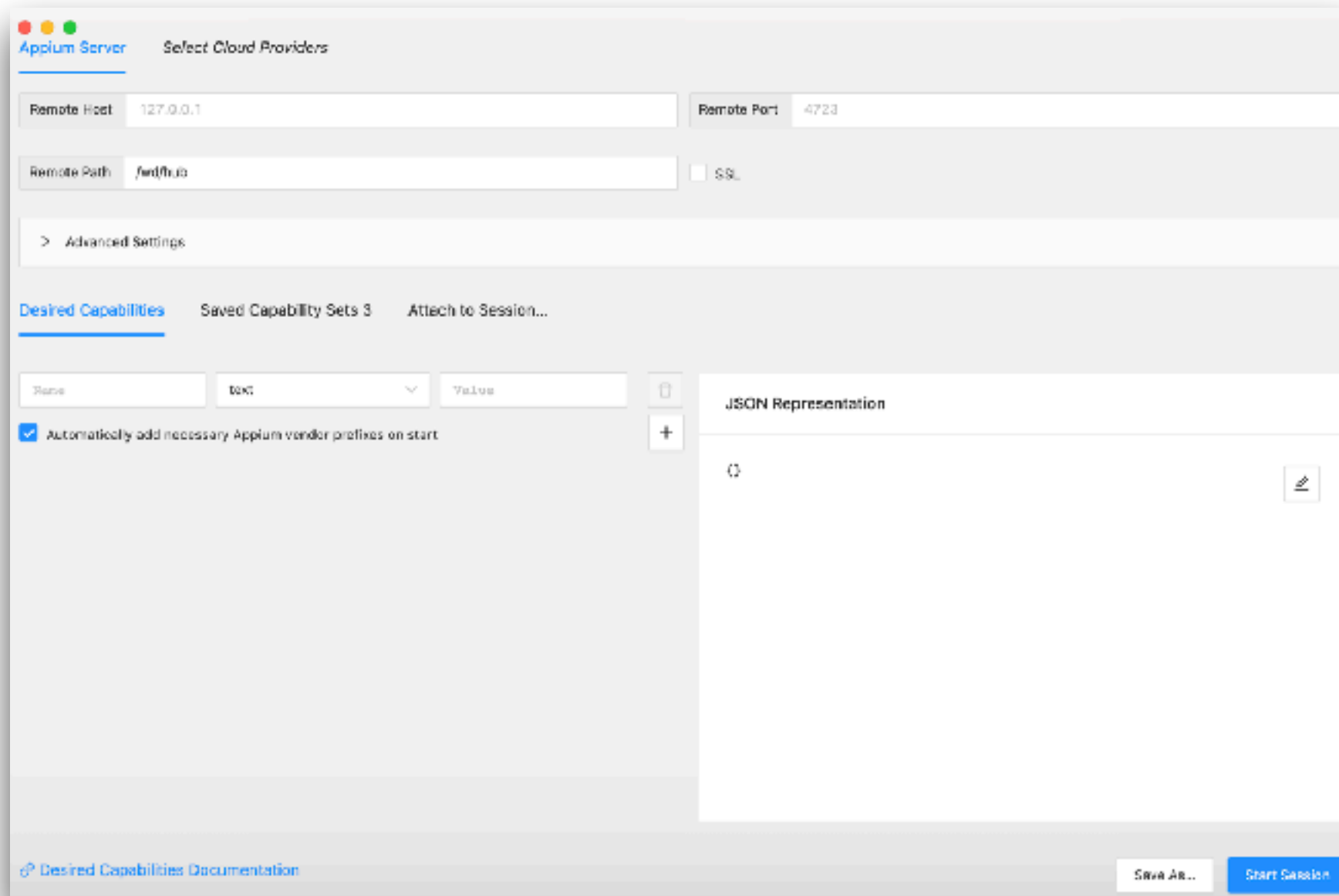


Appium Inspector



Appium Inspector

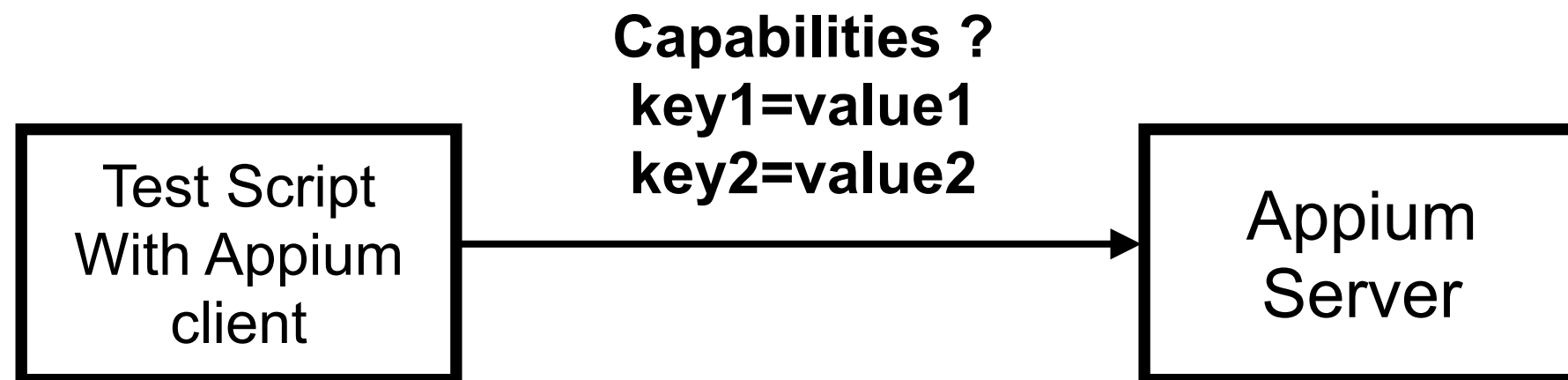
GUI inspector for mobile apps



<https://github.com/appium/appium-inspector>



Create a session with capabilities



<http://appium.io/docs/en/writing-running-appium/caps/index.html>



Required capabilities

Name	Description
platformName	Platform to automate (iOS, Android)
platformVersion	Version of platform
deviceName	Type of device to automate
app	Path to your app

<https://appium.io/docs/en/2.1/guides/caps/>



Session capabilities for android

```
{  
  "platformName": "Android",  
  "appium:automationName": "UiAutomator2",  
  "appium:appPackage": "com.demo.registerapp",  
  "appium:deviceName": "R58M36QKSJK"  
}
```

<https://github.com/appium/appium-uiautomator2-driver>



Appium Inspector (Android)

The screenshot displays the Appium Inspector interface for an Android application. On the left, a preview of the app shows a login screen with a purple header, a 'Sign In' title, and input fields for 'Username' and 'Password', followed by a 'Login' button. The top navigation bar includes icons for session control and tabs for 'Source', 'Commands', 'Gestures', 'Recorder', and 'Session Information'. The 'Source' tab is active, showing the XML source code of the app. The 'Selected Element' panel on the right provides details for the selected 'Login' button, including its accessibility id, id, class name, and various selectors. Below this, a 'Box Model' diagram shows the element's dimensions and coordinates. At the bottom, a table lists the element's attributes and their values.

App Source

```
<android.widget.FrameLayout>
  <android.widget.LinearLayout>
    <android.widget.FrameLayout resource-id="android:id/content">
      <android.widget.FrameLayout>
        <android.view.View>
          <android.view.View>
            <android.view.View>
              <android.view.View>
                <android.view.View content-desc="login_app_bar_title">
                  <android.view.View>
                    <android.view.View content-desc="login_title_text" resource-id="login_title_text2">
                      <android.widget.EditText resource-id="login_username_field2">
                        <android.widget.EditText resource-id="login_password_field2">
                          <android.widget.Button content-desc="Login" resource-id="login_submit_button">
```

Selected Element

Find By	Selector
accessibility id	Login
id	login_submit_button
class name	android.widget.Button
-android uiautomator	new UiSelector().resourceId("login_submit_button") (docs)
xpath	//android.widget.Button[@content-desc="Login"]

Box Model

126 954

(63, 1565) (540, 1628) (1017, 1565) (63, 1691) (1017, 1691)

Attribute	Value
elementId	00000000-0000-0007-0000-001100000000



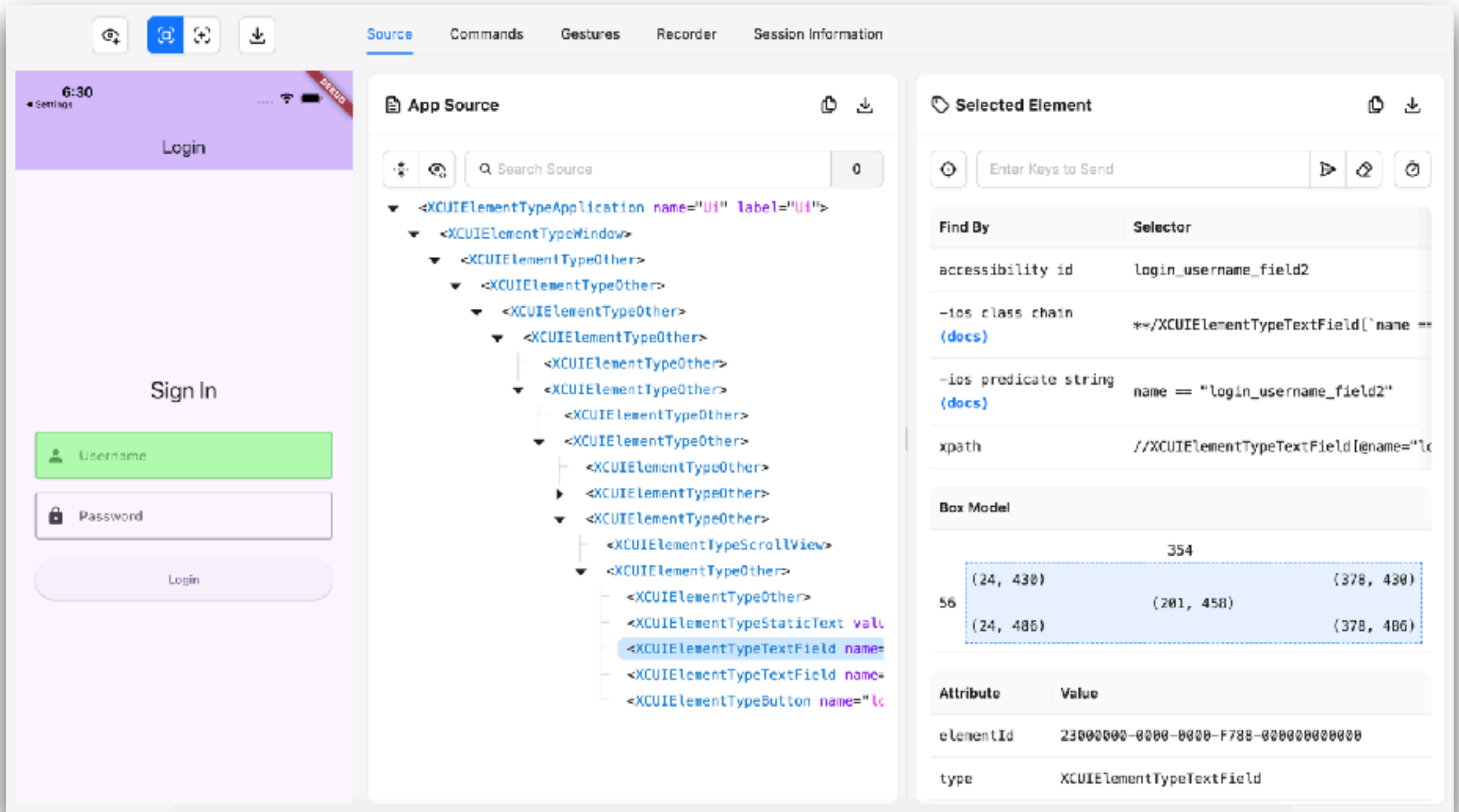
Session capabilities for iOS

```
{  
  "platformName": "iOS",  
  "appium:deviceName": "iPhone 17",  
  "appium:platformVersion": "26.2",  
  "appium:automationName": "XCUITest",  
  "appium:bundleId": "com.example.ui",  
  "appium:connectionRetryTimeout": 60000,  
  "appium:noReset": true  
}
```

<https://appium.github.io/appium-xcuitest-driver/latest/>



Appium Inspector (iOS)



Appium with Flutter App

Android App

iOS App



Finding and Using Elements

Isolated from UI !!

Android App

iOS App



Locator Strategies

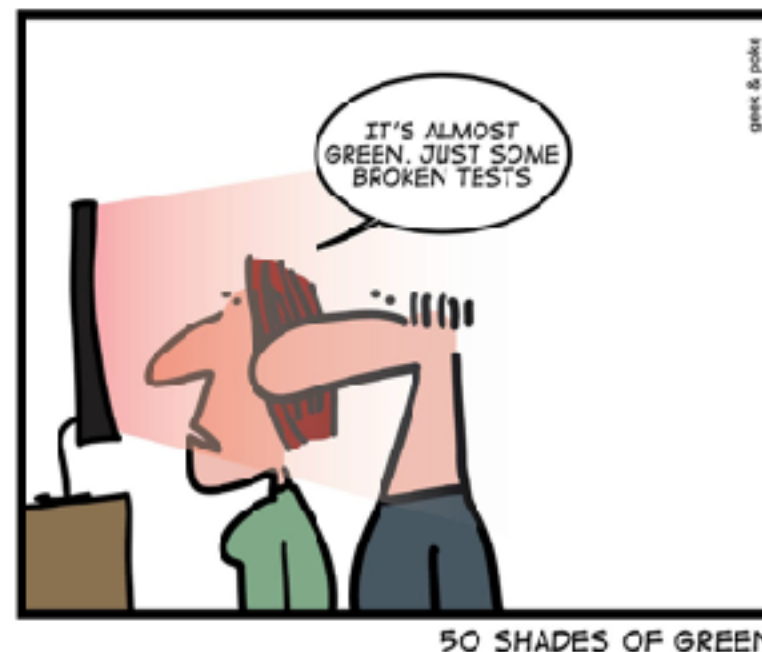
Strategy	Description
Accessibility ID	iOS = accessibility-id Android = content-desc
ID	iOS = name Android = resource-id
Name	Name of element
XPath	Pattern of element in XML (not recommended) Absolute path vs Relative path ?
Android UIAutomator	Android only
iOS Predicate String	iOS only



Good Locators

Unique
Descriptive
Resilient

Shorter in length (maintainability)



Flutter App Locator

Key (default)

Text

Semantic identifier

Tooltip

Type



Working with Semantics widget

Use identifier in Flutter 3.19+

Android App

iOS App

<https://blog.flutter.dev/whats-new-in-flutter-3-19-58b1aae242d2>



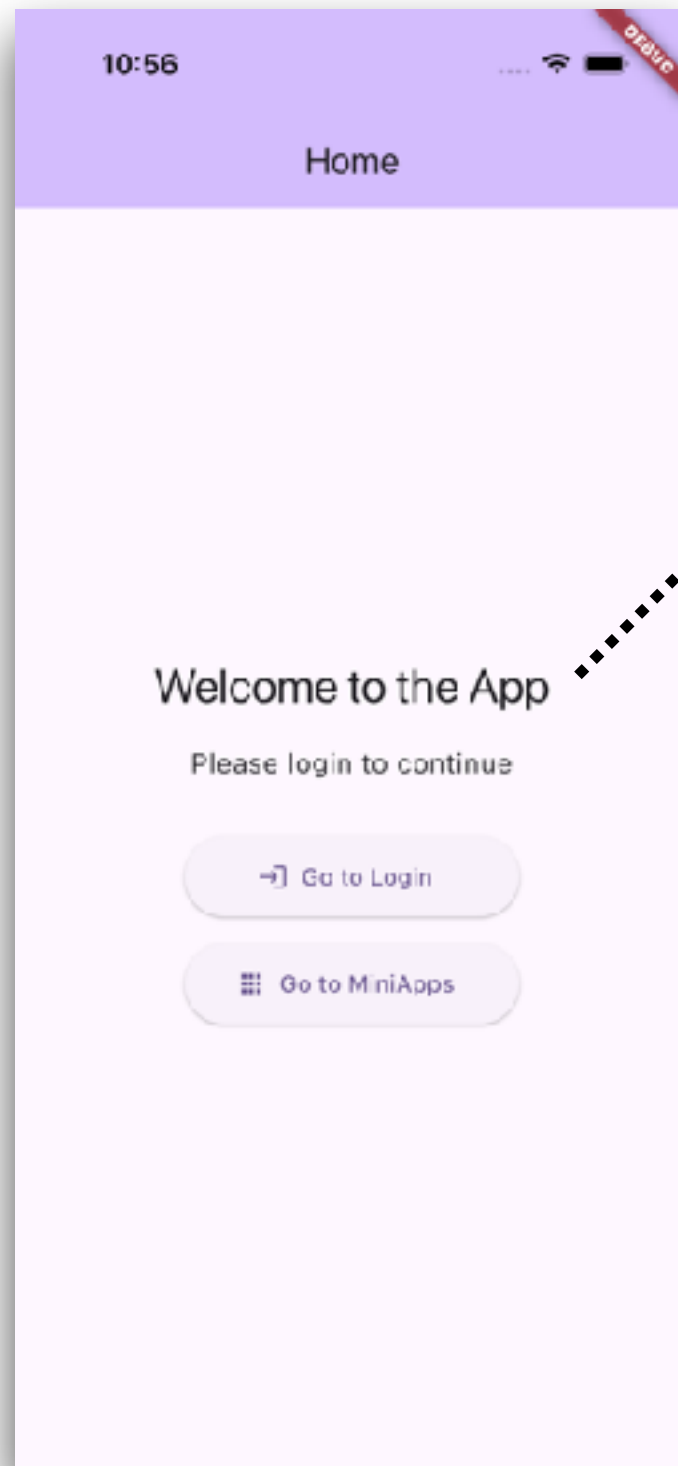
SemanticsProperties accessibility identifier

A new accessibility identifier in `SemanticsProperties` provides an identifier for the semantic node in the native accessibility hierarchy. On Android, it appears in the accessibility hierarchy as `resource-id`. On iOS, this sets `UIAccessibilityElement.accessibilityIdentifier`. We want to thank community member [@bartekpacia](#) for this change, which spanned the engine and framework.

<https://blog.flutter.dev/whats-new-in-flutter-3-19-58b1aae242d2>



Text (iOS app)



Flutter code

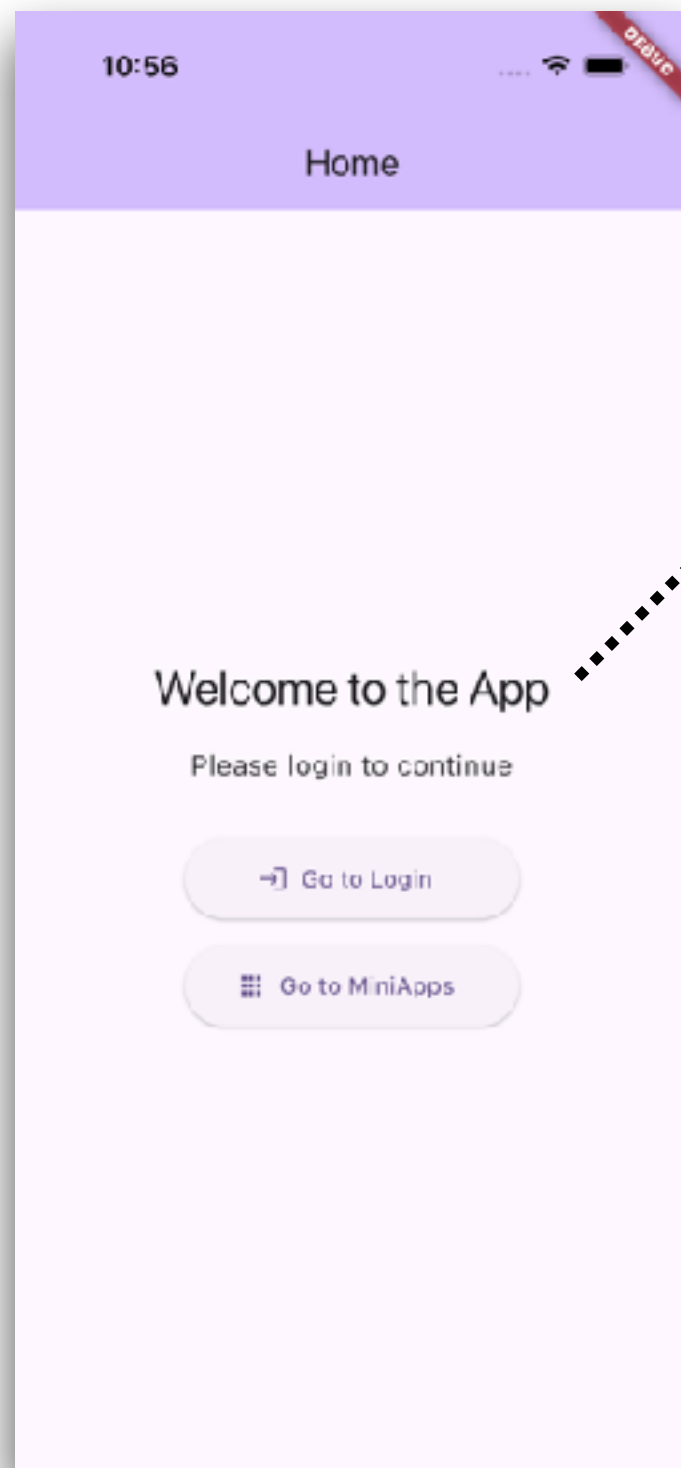
```
Text(  
  'Welcome to the App',  
  key: const Key('home_welcome_text'),  
  semanticsIdentifier: 'home_welcome_text2',  
)
```

Appium inspector (accessibility, text)

Find By	Selector
accessibility id	home_welcome_text2
-ios class chain (docs)	**/XCUIElementTypeStaticText[`name =
-ios predicate string (docs)	name == "home_welcome_text2"
xpath	//XCUIElementTypeStaticText[@name="I



Text (Android app)



Flutter code

```
Text(  
  'Welcome to the App',  
  key: const Key('home_welcome_text'),  
  semanticsIdentifier: 'home_welcome_text2',  
)
```

Appium inspector (resource-id, content-desc)

Find By	Selector
accessibility id	Welcome to the App
id	home_welcome_text2
-android uiautomator (docs)	new UiSelector().resourceId("home_wel
xpath	//android.view.View[@content-desc="We



Button (iOS app)

Flutter code

```
Semantics(  
  identifier: 'home_login_button',  
  label: 'Go to Login',  
  excludeSemantics: true,  
  child: ElevatedButton.icon(  
    key: const Key('home_login_button'),  
    label: const Text('Go to Login'),  
  ),  
)
```

Appium inspector (accessibility, value)

Find By	Selector
accessibility id	home_login_button
-ios class chain (docs)	**/XCUIElementTypeStaticText['name =
-ios predicate string (docs)	name == "home_login_button"
xpath	//XCUIElementTypeStaticText[@name="f



Button (Android app)

Flutter code

```
Semantics(  
  identifier: 'home_login_button',  
  label: 'Go to Login',  
  excludeSemantics: true,  
  child: ElevatedButton.icon(  
    key: const Key('home_login_button'),  
    label: const Text('Go to Login'),  
  ),  
)
```

Appium inspector (resource-id, content-desc)

Find By	Selector
accessibility id	Go to Login
id	home_login_button
-android uiautomator (docs)	new UiSelector().resourceId("home_log
xpath	//android.view.View[@content-desc="Go



ListView (iOS app)

Flutter code

```
return ListView.builder(  
  key: const Key('miniapps_list'),  
  itemCount: _miniApps!.length,  
  itemBuilder: (context, index) {  
    final miniApp = _miniApps![index];  
    return Semantics(  
      identifier: 'miniapp_list',  
      container: true,  
      child: Card(  
        key: Key('miniapp_card_${miniApp.id}'),  

```

Appium inspector (accessibility)

```
<XCUIElementTypeScrollView>  
└─ <XCUIElementTypeOther>  
  └─ <XCUIElementTypeOther>  
    └─ <XCUIElementTypeOther>  
      └─ <XCUIElementTypeOther name="miniapp_list">  
        └─ <XCUIElementTypeButton name="miniapp_name_1" label="1\nhttps://miniapp1.example.com">  
          └─ <XCUIElementTypeOther>  
            └─ <XCUIElementTypeOther>  
              └─ <XCUIElementTypeOther name="miniapp_list">  
                └─ <XCUIElementTypeButton name="miniapp_name_2" label="2\nhttps://miniapp2.example.com">
```



ListView (Android app)

Flutter code

```
return ListView.builder(  
  key: const Key('miniapps_list'),  
  itemCount: _miniApps!.length,  
  itemBuilder: (context, index) {  
    final miniApp = _miniApps![index];  
    return Semantics(  
      identifier: 'miniapp_list',  
      container: true,  
      child: Card(  
        key: Key('miniapp_card_${miniApp.id}'),  

```

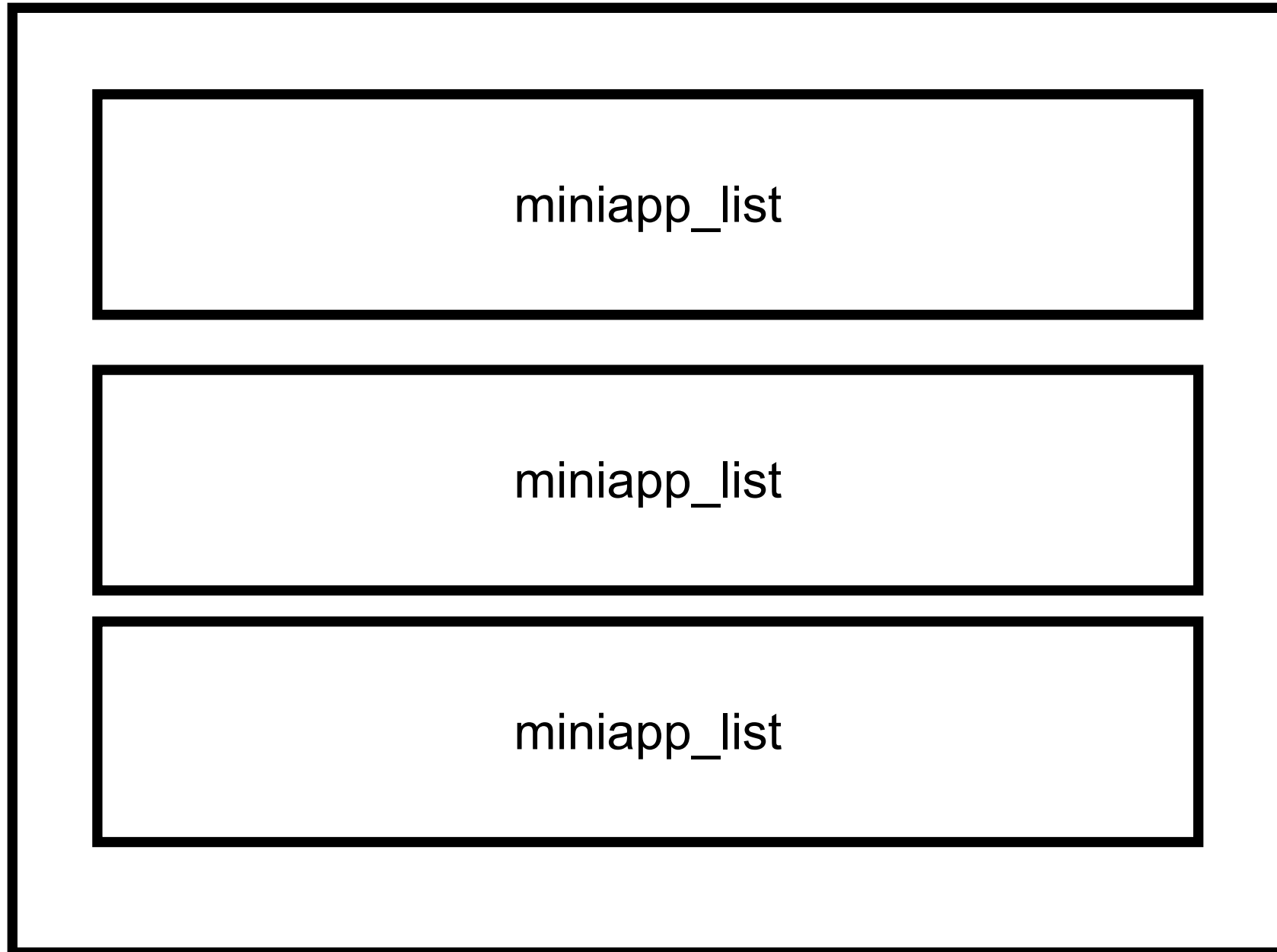
Appium inspector (resource-id)

```
<android.view.View>  
└─ <android.view.View resource-id="miniapp_list">  
    └─ <android.widget.Button content-desc="1\nMiniApp  
1\nhttps://miniapp1.example.com" resource-  
        id="miniapp_name_1">  
└─ <android.view.View>  
    └─ <android.view.View resource-id="miniapp_list">
```



ListView

Size = 3



Locator and Value ?

Flutter semantic	iOS	Android
Text	Accessibility-id Text	Resource-id Content-desc
Button	Accessibility-id Text	Resource-id Content-desc
TextField	Accessibility-id Value	Resource-id Value
ListView	Accessibility-id	Resource-id



Record from Appium Inspector

The screenshot displays the Appium Inspector interface. On the left, a mobile app is shown with a 'Home' screen. The app has a purple header with the time '9:03' and a 'Home' label. Below the header, there is a 'Welcome to the App' message and a 'Please login to continue' prompt. There are three buttons: 'Go to Login', 'Go to MiniApps', and 'Go to Products'. The 'Go to Login' button is highlighted with a red dashed box and a red circle with the number '1' next to it, with the text 'Start recording' above it.

In the center, the 'App Source' panel shows the XML source code of the app. The code is a tree structure of `<XCUIElementType...` elements. The 'Go to Login' button is highlighted with a red dashed box and a red circle with the number '2' next to it, with the text 'Interact with elements' above it.

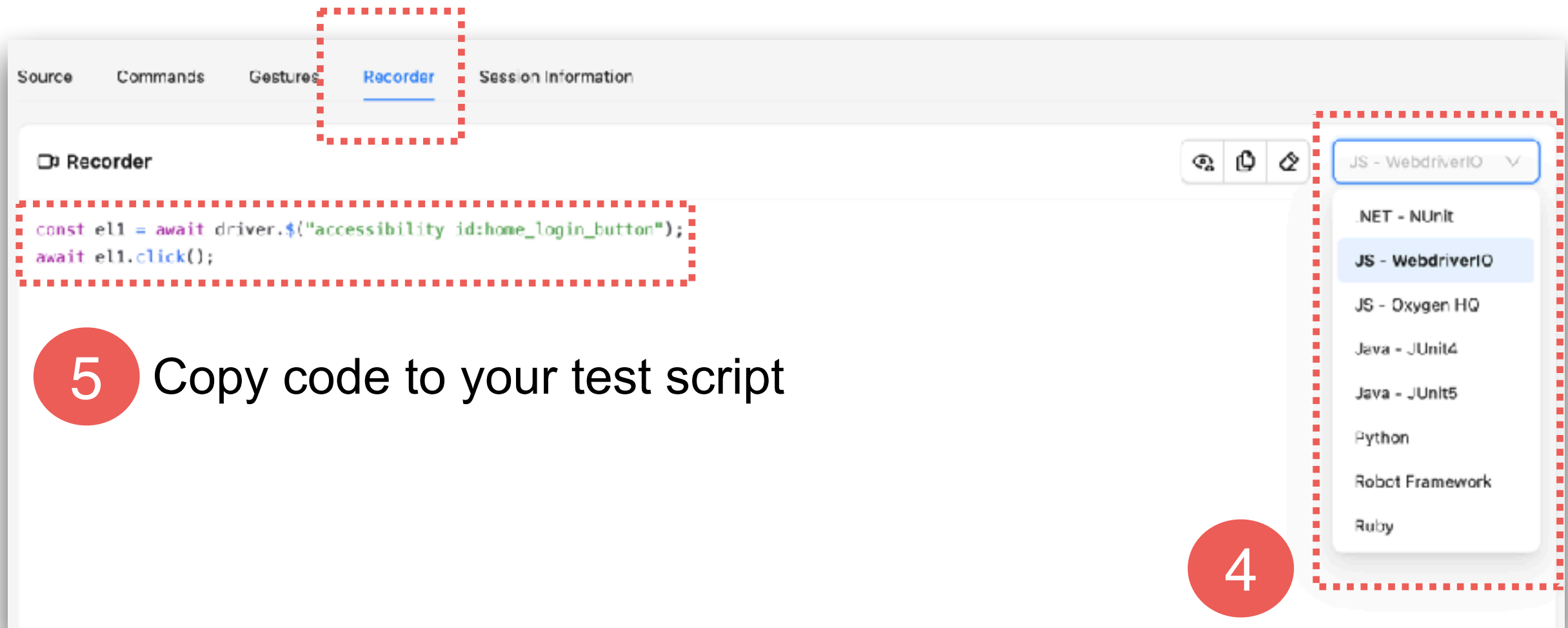
On the right, the 'Selected Element' panel shows the details of the selected element. It includes a 'Find By' section with a 'Selector' field containing the XPath `//XCUIElementTypeStaticText[@name='home_login_button']`. Below this is a 'Box Model' section showing the bounding box of the element. At the bottom is an 'Attribute' table.

Attribute	Value
elementId	11000000-0000-0000-5249-010000000000
type	XCUIElementTypeStaticText
value	Go to Login
name	home_login_button
label	Go to Login
enabled	true



Record from Appium Inspector

3 Go to tab Recorder



5 Copy code to your test script

Choose JS-WebdriverIO

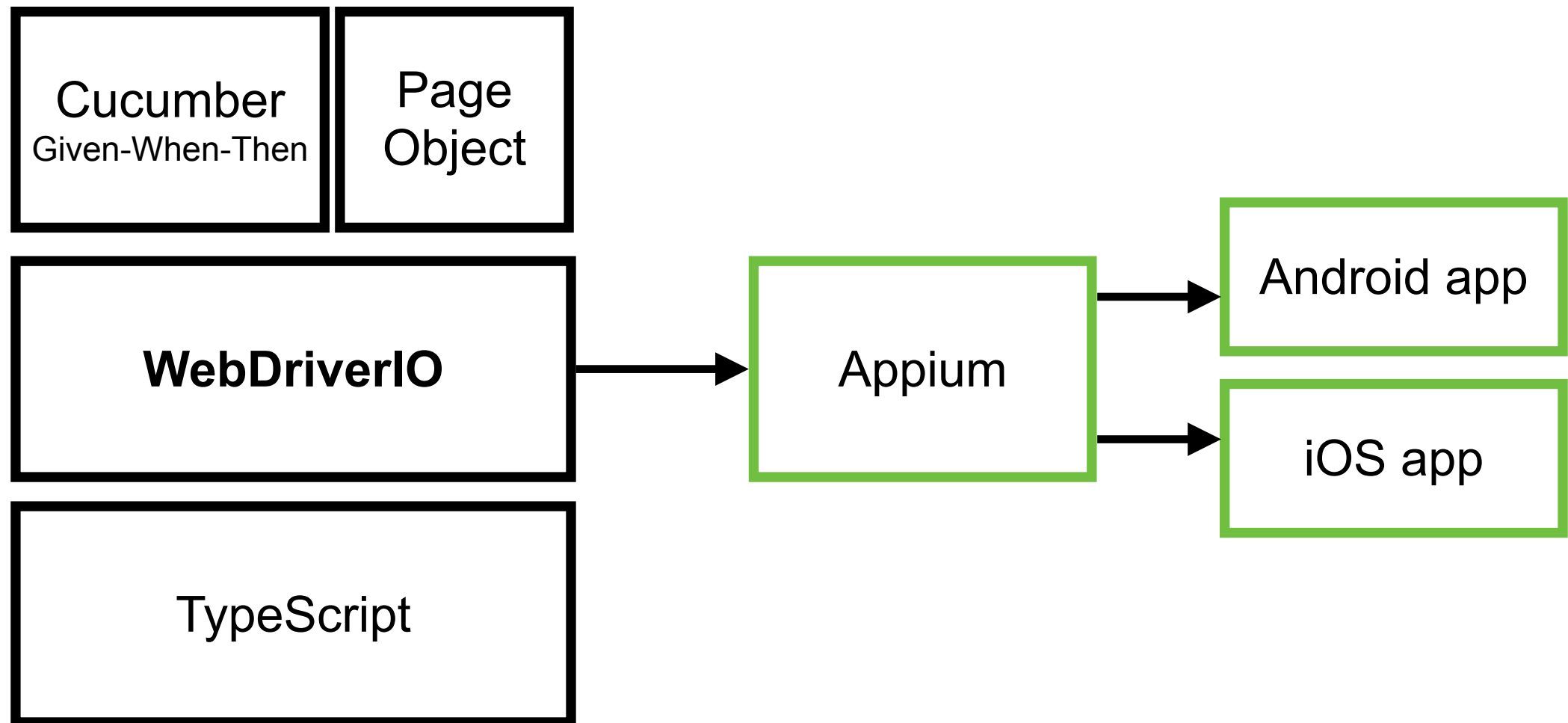
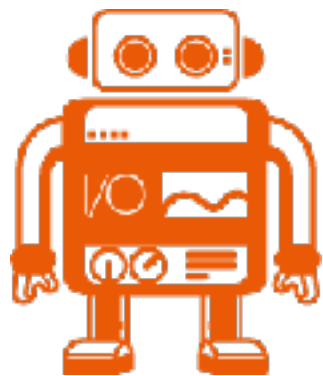


Write Test Cases



Create Test Project

\$npm init wdio@latest .



<https://webdriver.io/docs/gettingstarted>



Structure of project

File/Folder name	Description
wdio.conf.ts	ไฟล์ config หลักของ WebDriverIO
test/specs	จัดเก็บ test case ในรูปแบบของ Mocha
test/pageobjects	จัดเก็บ action, assert ต่าง ๆ ของแต่ละ page ตามแนวคิดของ POM
test/utils	จัดเก็บไฟล์ util หรือตัวช่วยของการทดสอบ ในตัวอย่างจะเป็นตัวจัดการ locator ของ ios/android

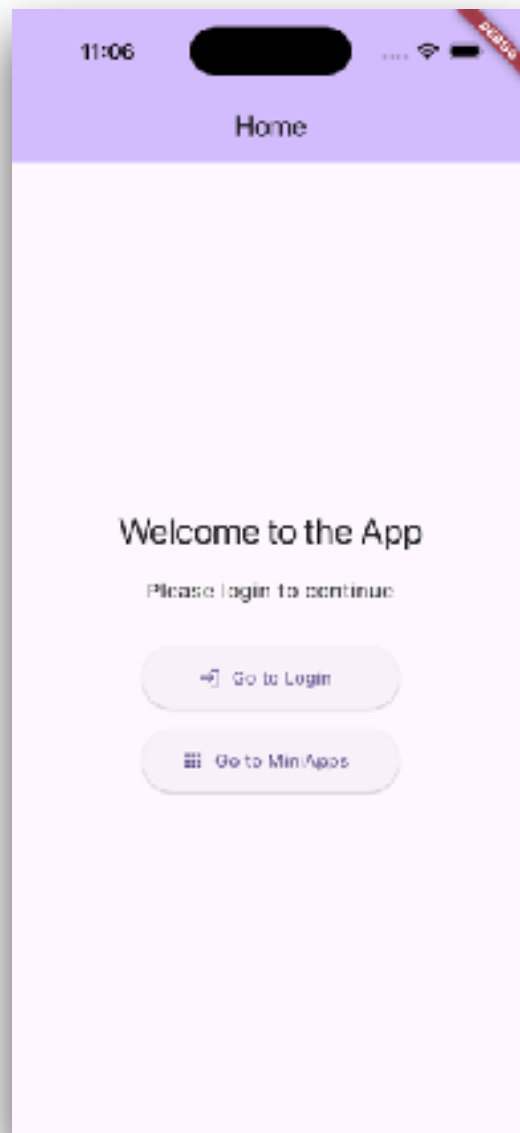


Mobile App for Testing

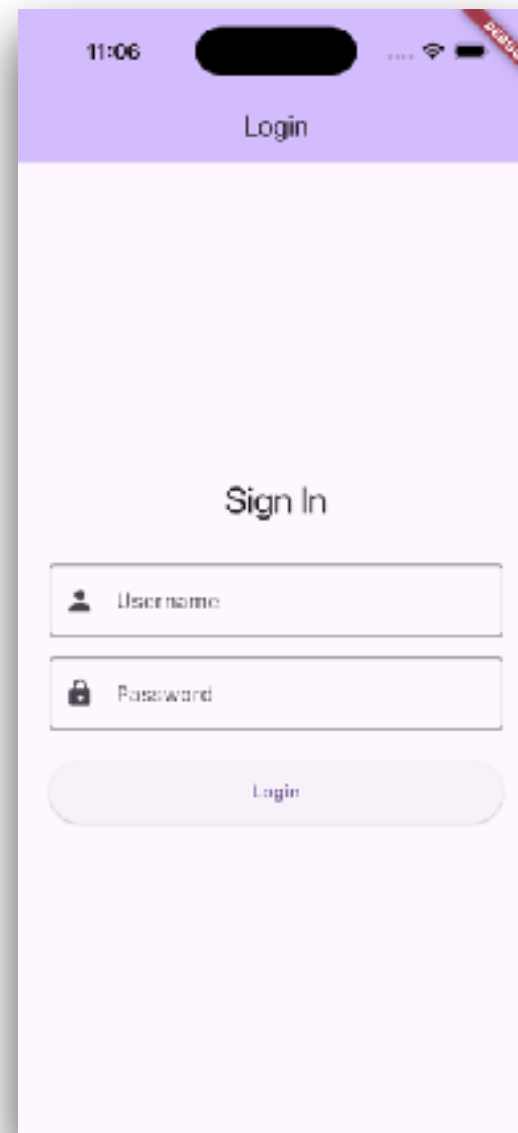


Flow of Mobile app (1)

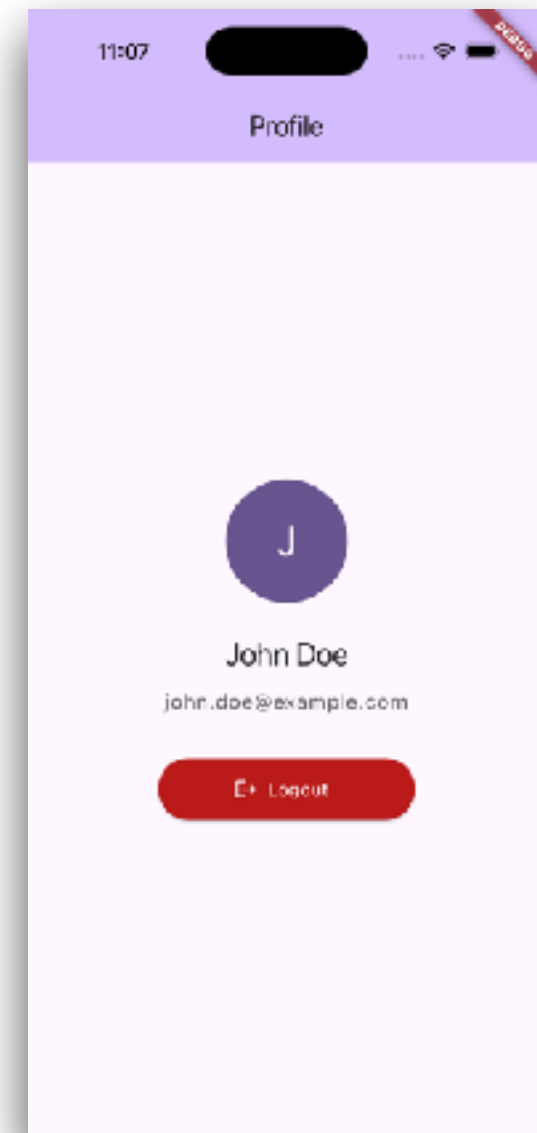
Home



Login

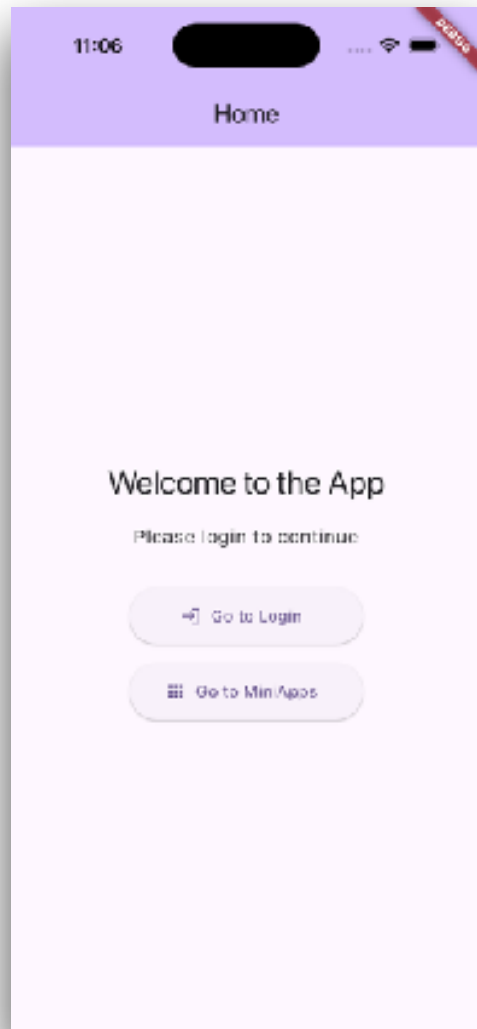


Profile

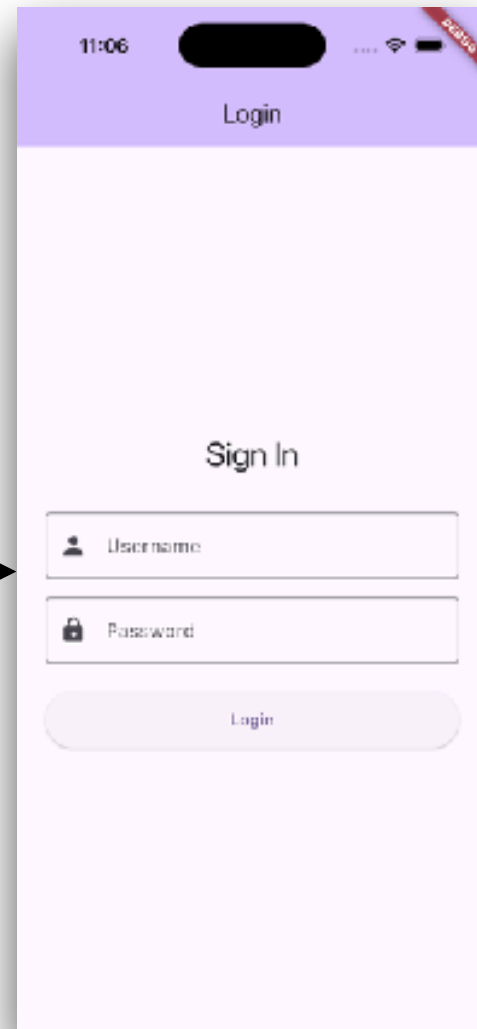


Flow of Mobile app (2)

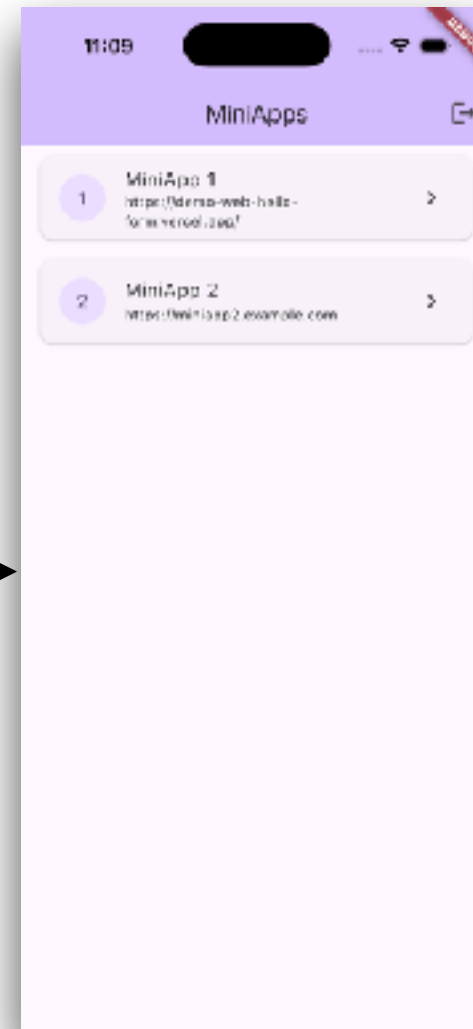
Home



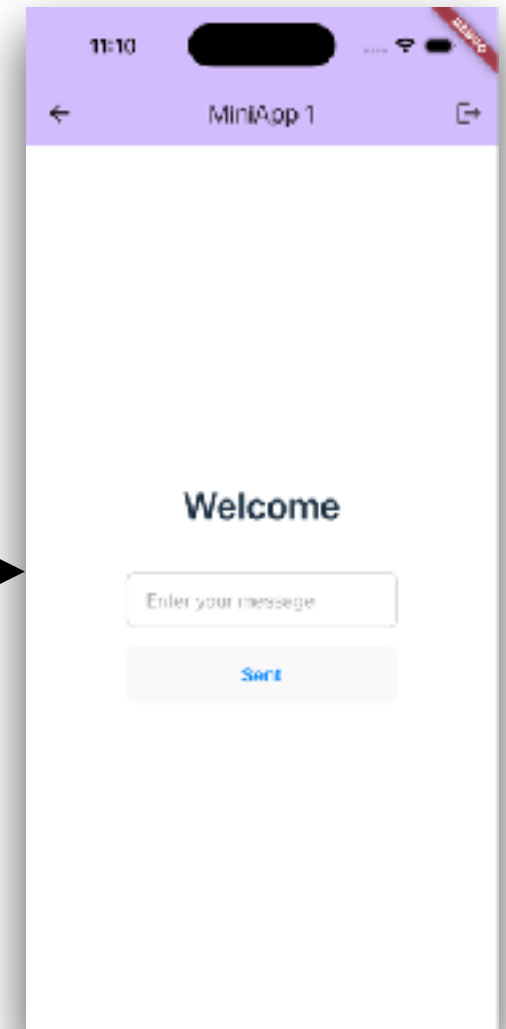
Login



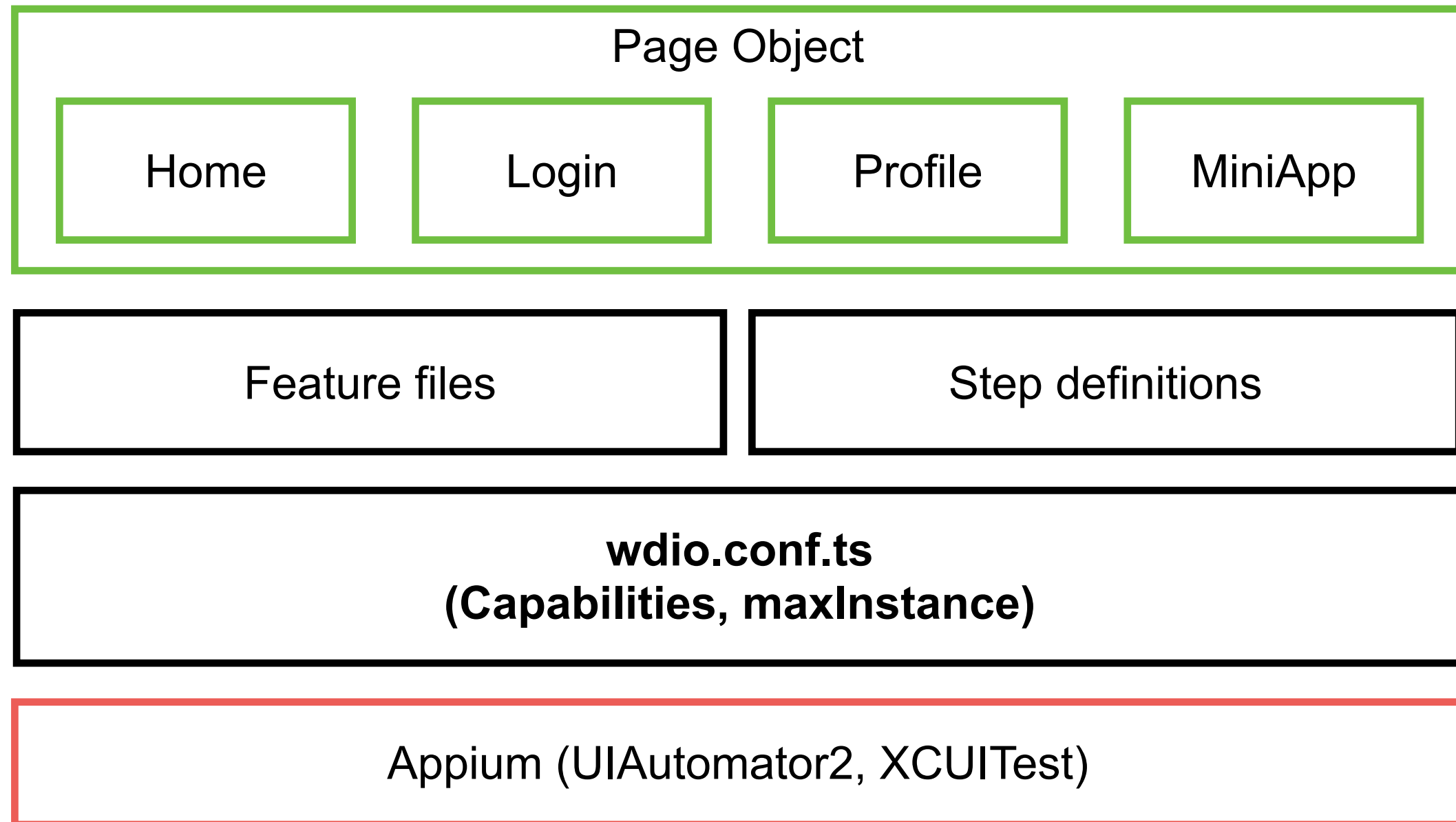
MiniAppList



WebView



Test Project Structure



https://github.com/up1/workshop-mobile-testing-with-typescript/tree/main/appium_testing



Test strategy ?

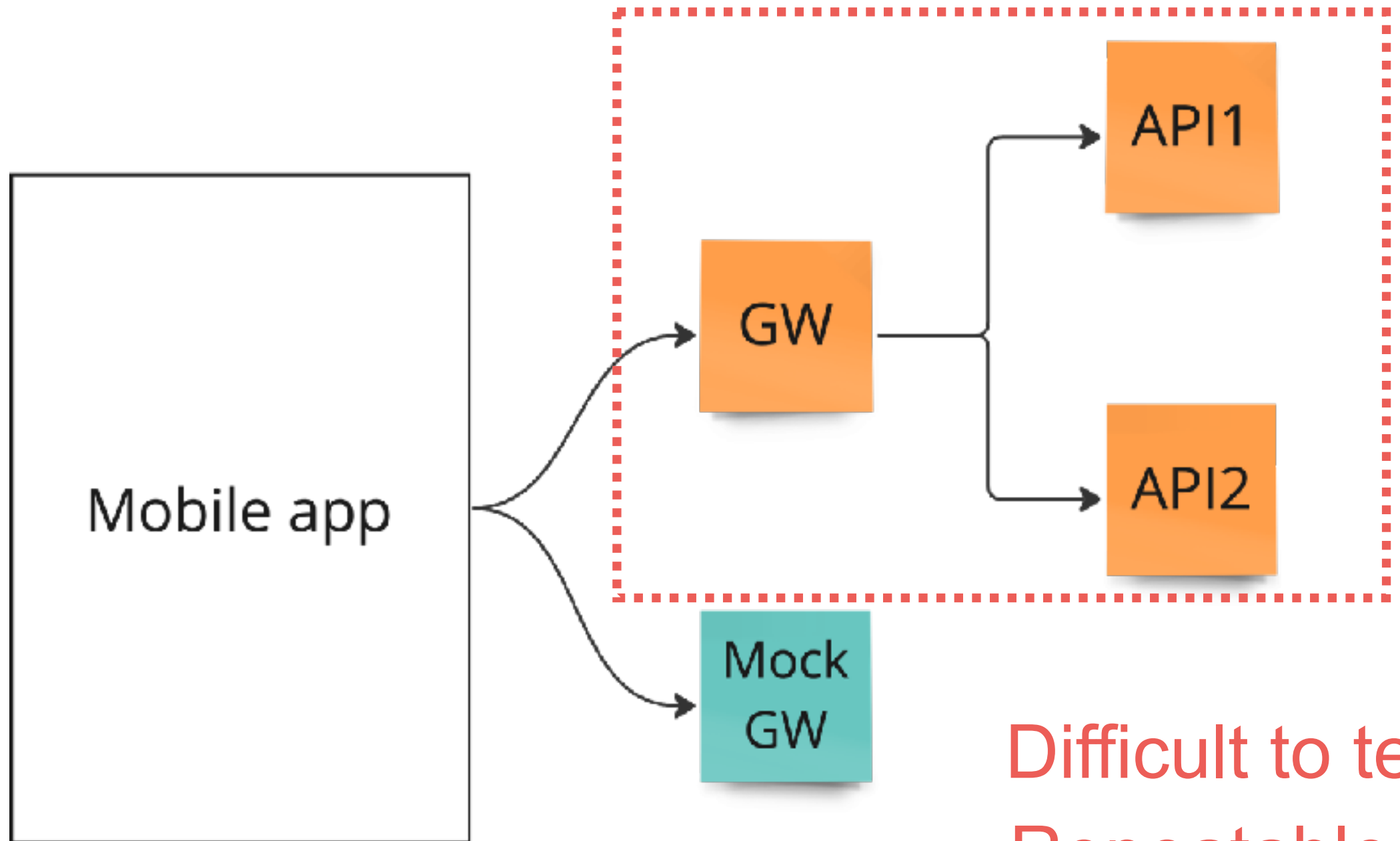


Good Tests ?

Fast
Isolate
Repeatable
Self-verify
Timely



Integration Testing



Difficult to test
Repeatable
Reliable



Q/A

